Q1.1: Add Receptionist Class (MOST COMMON QUESTION)

Question: Create a Receptionist class that extends HotelStaff with an additional field languagesSpoken (String). Update the system to allow adding receptionists through the console menu and display them correctly in the GUI table.

SOLUTION STEPS:

STEP 1: Create Receptionist.java (NEW FILE)

```
package com.mycompany.hotelmanagementsystem;


public class Receptionist extends HotelStaff {
  private String languagesSpoken;


  public Receptionist(String name, String surname) {
    super(name, surname);
  }


  // Setter and Getter methods
  public void setLanguagesSpoken(String languagesSpoken) {
    this.languagesSpoken = languagesSpoken;
  }


  public String getLanguagesSpoken() {
    return languagesSpoken;
  }


  @Override
  public String toString() {
    return super.toString() + " Receptionist - Languages: " + languagesSpoken;
  }
}
```

STEP 2: Update WestminsterHotelManager.java - addHotelStaff() method

FIND THIS SECTION:

```
System.out.println("Press 1 if you want to add a Manager");
System.out.println("Press 2 if you want to add a Housekeeper");
```

CHANGE TO:

```
System.out.println("Press 1 if you want to add a Manager");
System.out.println("Press 2 if you want to add a Housekeeper");
System.out.println("Press 3 if you want to add a Receptionist");  // ADD THIS LINE
```

FIND THIS SECTION (switch statement):

```
switch (choiceStaff) {
   case 1:
      // Manager code
      break;
   case 2:
      // HouseKeeper code
      break;
}
```

ADD THIS CASE BEFORE THE CLOSING }:

```
case 3:
   // It is a Receptionist
   System.out.println("Enter the languages spoken");
   String languages = s.nextLine();

   Receptionist receptionist = new Receptionist(name, surname);
   receptionist.setLanguagesSpoken(languages);
   receptionist.setDateOfBirth(date);
   receptionist.setStaffID(staffID);
   this.addStaffToList(receptionist);
```

```
    break;
```

STEP 3: Update HotelTableModel.java - getValueAt() method

FIND THIS SECTION:

```
} else if (columnIndex == 3) {
    if (list.get(rowIndex) instanceof Manager) {
        temp = "Manager";
    } else if (list.get(rowIndex) instanceof HouseKeeper) {
        temp = "HouseKeeper";
    }
}
```

CHANGE TO:

```
} else if (columnIndex == 3) {
    if (list.get(rowIndex) instanceof Manager) {
        temp = "Manager";
    } else if (list.get(rowIndex) instanceof HouseKeeper) {
        temp = "HouseKeeper";
    } else if (list.get(rowIndex) instanceof Receptionist) {  // ADD THIS
        temp = "Receptionist";
    }
}
```

Q1.2: Add Chef Class (VARIATION)

Question: Create a Chef class that extends HotelStaff with an additional field speciality (String). Modify the addHotelStaff() method to include option 3 for adding chefs. Update the table model to display "Chef" in the Role column.

SOLUTION STEPS:

STEP 1: Create Chef.java (NEW FILE)

```
package com.mycompany.hotelmanagementsystem;
```

```
public class Chef extends HotelStaff {
  private String speciality;

  public Chef(String name, String surname) {
    super(name, surname);
  }

  // Setter and Getter methods
  public void setSpeciality(String speciality) {
    this.speciality = speciality;
  }

  public String getSpeciality() {
    return speciality;
  }

  @Override
  public String toString() {
    return super.toString() + " Chef - Speciality: " + speciality;
  }
}
```

STEP 2: Update WestminsterHotelManager.java

ADD TO MENU:

```
System.out.println("Press 3 if you want to add a Chef");
```

ADD TO SWITCH STATEMENT:

```
case 3:
  // It is a Chef
```

```
System.out.println("Enter the chef's speciality");
String speciality = s.nextLine();

Chef chef = new Chef(name, surname);
chef.setSpeciality(speciality);
chef.setDateOfBirth(date);
chef.setStaffID(staffID);
this.addStaffToList(chef);
break;
```

STEP 3: Update HotelTableModel.java

ADD TO getValueAt() method:

```
} else if (list.get(rowIndex) instanceof Chef) {
   temp = "Chef";
```

Q1.3: Add SecurityGuard Class (VARIATION)

Question: Create a SecurityGuard class that extends HotelStaff with an additional field shiftTime (String). Add this new staff type to the menu system and ensure it appears correctly in the staff table.

SOLUTION STEPS:

STEP 1: Create SecurityGuard.java (NEW FILE)

```
package com.mycompany.hotelmanagementsystem;

public class SecurityGuard extends HotelStaff {
   private String shiftTime;

   public SecurityGuard(String name, String surname) {
      super(name, surname);
   }
```

```
    // Setter and Getter methods
    public void setShiftTime(String shiftTime) {
        this.shiftTime = shiftTime;
    }

    public String getShiftTime() {
        return shiftTime;
    }

    @Override
    public String toString() {
        return super.toString() + " Security Guard - Shift: " + shiftTime;
    }
}
```

STEP 2: Update WestminsterHotelManager.java

ADD TO MENU:

```
System.out.println("Press 3 if you want to add a Security Guard");
```

ADD TO SWITCH STATEMENT:

```
case 3:
    // It is a Security Guard
    System.out.println("Enter the shift time (Day/Night/Evening)");
    String shift = s.nextLine();

    SecurityGuard guard = new SecurityGuard(name, surname);
    guard.setShiftTime(shift);
    guard.setDateOfBirth(date);
    guard.setStaffID(staffID);
    this.addStaffToList(guard);
```

```
    break;
```

STEP 3: Update HotelTableModel.java

ADD TO getValueAt() method:

```
} else if (list.get(rowIndex) instanceof SecurityGuard) {
    temp = "Security Guard";
```

Q1.4: Fix Guest toString() Bug (EASY POINTS)

Question: The Guest class has a bug in the toString() method that displays memory addresses instead of guest names. Fix this bug and add the missing getName() and getSurname() getter methods that are currently absent from the class.

SOLUTION STEPS:

STEP 1: Update Guest.java - Add Missing Getters

FIND THE EXISTING GETTERS SECTION and ADD:

```
// ADD THESE MISSING GETTER METHODS
public String getName() {
    return name;
}

public String getSurname() {
    return surname;
}
```

STEP 2: Fix toString() Method

FIND THIS BUGGY CODE:

```
@Override
public String toString() {
    return super.toString() + " Guest - Room number: " + roomNumber + ", Nights stayed: " + nightsStayed;
}
```

REPLACE WITH:

```
@Override
```

```
public String toString() {
   return name + " " + surname + " - Room number: " + roomNumber + ", Nights stayed: " + nightsStayed;
}
```

Q1.5: Make Info Button Functional (GUI FUNDAMENTALS)

Question: The "Info" button in HotelTableGUI currently does nothing. Add functionality so that when clicked, it displays detailed information about the selected staff member in a popup dialog. Handle the case when no row is selected.

SOLUTION STEPS:

STEP 1: Update HotelTableGUI.java - Add Import

ADD THIS IMPORT AT THE TOP:

```
import javax.swing.JOptionPane;
```

STEP 2: Update Button Creation Section

FIND THIS CODE:

```
// Add a button at the bottom
JButton button = new JButton("Info");


// Add the panel to the frame
add(scrollPane, BorderLayout.CENTER);
add(button, BorderLayout.SOUTH);
```

REPLACE WITH:

```
// Add a button at the bottom
JButton button = new JButton("Info");
button.addActionListener(e -> {
   int selectedRow = myTable.getSelectedRow();
   if (selectedRow >= 0) {
      // Convert view row to model row (important for sorted tables)
      int modelRow = myTable.convertRowIndexToModel(selectedRow);
      HotelStaff staff = list.get(modelRow);
```

```
    JOptionPane.showMessageDialog(this, staff.toString(), "Staff Information",
JOptionPane.INFORMATION_MESSAGE);
  } else {
    JOptionPane.showMessageDialog(this, "Please select a staff member first!", "No
Selection", JOptionPane.WARNING_MESSAGE);
  }
});

// Add the panel to the frame
add(scrollPane, BorderLayout.CENTER);
add(button, BorderLayout.SOUTH);
```

Q1.6: Add Delete Staff Button (GUI EXTENSION)

Question: Add a "Delete Staff" button to the HotelTableGUI that removes the selected staff member from the list and updates the table display. Include confirmation dialog before deletion.

SOLUTION STEPS:

STEP 1: Update HotelTableGUI.java - Modify Layout

FIND THIS SECTION:

```
// Add a button at the bottom
JButton button = new JButton("Info");
```

REPLACE WITH:

```
// Create a panel for multiple buttons
JPanel buttonPanel = new JPanel();

// Add Info button (with existing functionality)
JButton infoButton = new JButton("Info");
infoButton.addActionListener(e -> {
  int selectedRow = myTable.getSelectedRow();
  if (selectedRow >= 0) {
```

```
      int modelRow = myTable.convertRowIndexToModel(selectedRow);
      HotelStaff staff = list.get(modelRow);
      JOptionPane.showMessageDialog(this, staff.toString(), "Staff Information",
JOptionPane.INFORMATION_MESSAGE);
    } else {
      JOptionPane.showMessageDialog(this, "Please select a staff member first!", "No
Selection", JOptionPane.WARNING_MESSAGE);
    }
});

// Add Delete button
JButton deleteButton = new JButton("Delete Staff");
deleteButton.addActionListener(e -> {
    int selectedRow = myTable.getSelectedRow();
    if (selectedRow >= 0) {
      int modelRow = myTable.convertRowIndexToModel(selectedRow);
      HotelStaff staff = list.get(modelRow);

      // Confirmation dialog
      int confirm = JOptionPane.showConfirmDialog(this,
        "Are you sure you want to delete " + staff.getName() + " " + staff.getSurname() + "?",
        "Confirm Deletion",
        JOptionPane.YES_NO_OPTION);

      if (confirm == JOptionPane.YES_OPTION) {
        list.remove(modelRow);
        tableModel.fireTableDataChanged(); // Refresh the table
        JOptionPane.showMessageDialog(this, "Staff member deleted successfully!",
"Deleted", JOptionPane.INFORMATION_MESSAGE);
      }
```

```
    } else {
        JOptionPane.showMessageDialog(this, "Please select a staff member to delete!", "No Selection", JOptionPane.WARNING_MESSAGE);
    }
});

// Add buttons to panel
buttonPanel.add(infoButton);
buttonPanel.add(deleteButton);
```

STEP 2: Update the Layout Addition

FIND:

```
add(scrollPane, BorderLayout.CENTER);
add(button, BorderLayout.SOUTH);
```

REPLACE WITH:

```
add(scrollPane, BorderLayout.CENTER);
add(buttonPanel, BorderLayout.SOUTH);
```

STEP 3: Add Required Import

ADD AT TOP:

```
import javax.swing.JPanel;
```